Helix Support Console

Client tickets answered by Support in minutes, without paging engineering

Life-sciences client support triage on Databricks: Lakeflow, Unity Catalog, Lakebase, MLflow + AI Functions, Genie and Databricks Apps.

Audience: VP Customer Success (executive sponsor), CTO, Support lead and SRE lead (domain owners).

Helix Clinical Cloud, its clients and all data in this deck are fictional and synthetic.

1. The outcome

A support analyst who gets "site-204 can't upload scans" now opens **one app** and, within a minute, sees:

- the client's health by site and service, served from **Lakebase** (Postgres);
- whether the **ML model** flagged it, and when it started;
- an **AI incident brief** with likely cause, next step and a **ready-to-send client reply**;
- an **Ask Genie** box for follow-up questions in plain English;
- a **triage note** saved back to Lakebase for hand-off.

Proven in the prototype (measured)	Result
Injected client incidents found by the anomaly model	**3 of 3** (recall 1.0, ROC AUC 0.9999)
Incidents with an AI brief and client reply	**10** (2 × SEV1, 6 × SEV2, 2 × SEV3)
Logs ingested end-to-end (Fluent Bit → Zerobus → Delta)	**1.64 M rows, 1.8 GB in 10 min, 0 failed flushes**
Other-tenant rows exposed to Support	**0** (row filter; 196 k isolated-tenant rows hidden)
Genie questions answered with correct SQL	**5 of 5** sample questions + app question

2. The problem today

Helix runs imaging-archive, DICOM routing, trial-data upload and eConsent for pharma and CRO clients on Kubernetes.

- Logs go **disk → Vector → S3 → batch job → one untyped table**, more than **15 minutes late**.
- Support **cannot see application logs**. Every "my site's upload failed" ticket becomes an engineering escalation.
- Engineers grep raw logs per ticket; client communication waits on them.
- Missed clinical-trial milestones turn into **SLA breaches and service credits**.
- Scale: about **4.17 B log rows / 1.3 TB per day**, roughly **48 k lines per second** at peak.

3. KPIs: baseline and target

KPI	Baseline (today)	Target (90 days after rollout)	How it is measured
Mean time to triage a client ticket	~4 h (wait for engineer)	**< 15 min**	ticket open → first triage note in `triage_notes`
Engineering escalation rate	~70 % of app tickets	**< 25 %**	tickets with no engineering assignee
SLA breaches / service credits	quarterly credits on SEV1s	**-50 %**	SEV1 duration from `gold_incident_briefs`
Log freshness for Support	> 15 min, batch	**< 1 min** bronze, < 5 min gold	`ingest_time - event_time` in silver
Log pipeline infrastructure	Vector fleet + S3 staging + batch job	**Remove Vector and S3 hops**	Fluent Bit writes straight to Zerobus

Baselines are planning assumptions to be confirmed with Helix's ticketing data in week 1. The targets are what the pilot is meant to prove.

4. Value by audience

Executive sponsor: VP Customer Success / CTO

- Client-facing answers in minutes protect renewals with top pharma and CRO accounts.
- Fewer SLA credits; SEV1s are detected by the model rather than reported by the client.
- One governed platform instead of Vector + S3 + a separate log store: lower run cost and fewer moving parts.
- Isolation by client is enforced by Unity Catalog policy, not by convention, which keeps audits simple.

Domain owner: Support lead / SRE lead

- Support self-serves: health, anomalies, cause and reply text in one place.
- SRE gets fewer pages and a shared timeline (anomaly windows plus triage notes) when an escalation is needed.
- Genie answers ad-hoc questions ("worst site for uploads today?") without a dashboard request.
- Masked PII (coordinator email, study id) by default; privileged groups see full values.

5. Architecture

```
K8s app pods (synthetic generator)
   | container log files
   ▼
Fluent Bit 5.1 —OTLP gRPC—▶ Zerobus (Lakeflow Connect)
   ▼
UC bronze: OTEL v2 Delta table
   ▼  Lakeflow Spark Declarative Pipeline (serverless)
silver_otel_logs   typed, deduped, column masks, row filter
   ▼
gold_client_health
   ▼
MLflow IsolationForest → gold_client_anomalies
   ▼
ai_query briefs → gold_incident_briefs
              → gold_brief_inference (logged prompt/response)
   ▼
Lakebase SNAPSHOT sync (Postgres) + Genie Agent
   ▼
Databricks App "Support Console"
   └── writes triage_notes —▶ Lakebase
```

One Lakeflow Job runs: pipeline refresh → governance → train and score → AI briefs (from the model's flagged windows) → Lakebase sync.

6. What each Databricks capability does

Stage	Capability	What it does for Support
Ingest	**Lakeflow Connect: Zerobus** (OTLP gRPC)	Logs land in Delta seconds after they are written; no Vector or S3
Transform	**Lakeflow Spark Declarative Pipelines**	Typed silver with expectations and dedup; gold health in 5-minute windows
Govern	**Unity Catalog**	Column masks, client row filter, tags, grants, end-to-end lineage
Detect	**MLflow + UC model registry**	IsolationForest flags abnormal client/site/service windows
Explain	**AI Functions (`ai_query`)**	Structured brief: severity, cause, next step, client reply
Serve	**Lakebase** (synced tables + native table)	Millisecond app reads; triage notes written back
Ask	**Genie Agent**	Natural-language questions with trusted SQL and instructions
Act	**Databricks Apps**	Support Console with SSO and service-principal-scoped access

7. Demo flow (5 minutes)

1. **Client list**: Fabrikam CRO and Contoso Therapeutics are at the top with SEV1.
2. **Fabrikam CRO**: hourly error bars spike 17:40–19:15; site-204 trial-upload-api and dicom-router are at 44 % errors at peak.
3. **Brief**: "DICOM association rejected / upload timeout" with a recommended router fail-over and a client reply ready to send.
4. **Triage note**: the analyst logs SUP-4812 as *mitigated*; it is stored in Lakebase `support_app.triage_notes`.
5. **Ask Genie**: "Which Fabrikam CRO services had the worst error rate between 17:00 and 19:30?" returns trial-upload-api 5.9 % and dicom-router 5.7 %, with the SQL shown.
6. **Governance**: the same silver query shows `***@client.example` emails, and the isolated tenant is invisible.

8. Evidence from the prototype run

Measure	Value
Load test	600 s, 3 MB/s, 24 simulated hours, 2,879 HTTP 200 flushes, 0 failures
Bronze → silver	1,637,120 rows → 1,440,832 visible rows (isolated tenant filtered)
Gold health windows	26,010 client/site/service 5-minute windows
Anomaly model (v1 champion)	precision 0.53, recall 1.00, average precision 0.95, 130 windows flagged
AI briefs	10 incidents; both SEV1s match injected incidents; the archive slowdown was attributed to S3
Lakebase	3 synced tables online plus `triage_notes` ; app reads and writes verified
Genie	6 questions, all answered with correct SQL

The model is deliberately tuned for recall: missing a client incident costs far more than reviewing a false alarm. The briefs show severity, so analysts start with SEV1.

9. Rollout plan

Phase	Weeks	Scope	Exit criteria
Pilot	1–4	One region, 2 clients, Support pod of 5	Triage time < 30 min; analysts prefer the console
Expand	5–8	All US clients; Fluent Bit DaemonSet replaces Vector	Freshness < 1 min at 48 k lines/s; zero data loss
Scale	9–12	Non-US regions; ticketing integration (note ↔ ticket)	Escalations < 25 %; SLA-credit trend down
Operate	12+	Retrain model monthly; Genie benchmarks in CI	Model precision ≥ 0.7 on labeled incidents

Risks and mitigations: Zerobus token lifecycle (token broker sample in repo); false positives (severity triage, human in the loop); PII in logs (UC masks, tags, audit).

10. The ask

1. **Sponsor**: approve a 4-week pilot with two clients and one Support pod.
2. **Support lead**: name 5 pilot analysts and share a month of ticket data to confirm the baseline KPIs.
3. **SRE lead**: deploy the Fluent Bit DaemonSet to one cluster in parallel with Vector.
4. **Together**: agree on the KPI dashboard (triage time, escalation rate, SLA credits) and review it at week 4.

Repository, notebooks with outputs, and evidence: see `README.md` and `evidence/`.